Quantifying Confidence

Angeletos & Collard & Dellas *Econometrica* 2018

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PSE Macro reading group

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Motivation (1): concepts

"The state of confidence, as they term it, is a matter to which practical men always pay the closest and most anxious attention. But economists have not analysed it carefully and have been content, as a rule, to discuss it in general terms. In particular it has not been made clear that its relevance to economic problems comes in through its important influence on the schedule of the marginal efficiency of capital. There are not two separate factors affecting the rate of investment, namely, the schedule of the marginal efficiency of capital and the state of confidence. The state of confidence is relevant because it is one of the major factors determining the former, which is the same thing as the investment demand-schedule"

J.M. Keynes, *The General Theory of Employment, Interest and Money* (1936), chapter 12, section II.

Motivation (1): concepts

- Recessions are associated with self-confirming pessimism about the economic outlook, a weak state of confidence
- Full Information Rational Expectations assume an extreme form of expectation coordination: a perfect concensus about the current and future state of the economy is reached instantly
- Mutilple FIRE equilibria can model coordination failures but are (usually) unfit for quantitative analysis: which randomization over equilibria? Why?

Contribution: accomodate frictionnal expectation coordination as an autonomous driver of business cycles within a RE framework.

Motivation (2): technicalities

In a RE framework, the equilibrium object is $f(S_t)$. Fluctuations can be understood as:

1. A parametric change of f (shocks)
2. A locally unstable solution (endogenous b.c.)
3. A switch across the solution space (sunspots)

In the presence of higher-order uncertainty, the state space becomes very large and a numerical solution challenging (Townsend 1978):

- Agent i forms a belief about x
- Agents j forms a belief about i's belief about x
- Agent i forms a belief about j's belief about his belief about x ...

Contribution: accomodate something that looks like (3) in a setting that is like (1) while keeping a low dimensional S_t.

Overview of results

- Higher-order uncertainty is captured with heterogenous prior beliefs.
- Straightforward conceptual and numerical implementation via a simple enrichment of the state space
- Quantitative results:
  - Accounts for most of the volatility and co-movement of macro variables in both a RBC and a medium-scale DSGE model
  - Demand driven shocks outside of the inflation-output nexus
  - Waves of optimism and pessimism about the short-run, disconnected from "news shocks" about TFP

Conclusion: variations in "Sentiments" are the "Main Business Cycle Shock"

The basic model: setup

- Geography:
  - Continuum of islands i with a local labour and capital competitive markets to produce intermediate goods
  - Centralized competitive market for intermediate and final goods
- Timing: Each period is split into two stages:
  - t_1: N_i and K_i are chosen based on incomplete information about the choices made on other islands.
  - t_2: centralized market operates, N, K, Y are revealed, consumption/saving decisions are made.
- Standard preferences, CD technology for intermediate and final goods, TFP $\sim$ RW

Aggregation of local goods leads to a strategic complementarity across islands decisions which is key for imperfect communication to have an effect on equilibrium outcomes (// Lucas 1972):

$$\frac{p_{i,t}}{P_t} = \frac{Y_t}{y_{i,t}}$$

Information structure

Remove Common Knowledge of A_t in period t_1: each island is instead privately informed:

$$z_{i,t} = A_t + \varepsilon_{i,t}$$

Heterogenous priors: each island believes that its own signal is unbiased, but that the signals of all other islands are biased:

$$\varepsilon_{i,t} \sim \mathcal{N}(0, \sigma^2) \text{ and } \varepsilon_{j,t} \sim \mathcal{N}(\xi_t, \sigma^2), \ \forall j \neq i, \ \xi_t \sim AR(1)$$

Taken litteraly, this is a departure from RE: systematic bias in equilibrium expectations, no learning from past mistakes.

This is essentially a **modelling shortcut to allow for a tractable quantitative implementation of the model.**

Justification (1): a static version with random matching

Remove capital accumulation from the model, assume that trading takes places through random pairwise matching and reinterpret intermediate goods as final goods (Angeletos & La'O 2013).

Claim 1: the equilibrium can depend on a shock $\xi_t \perp\!\!\!\perp$ fundamentals:
Take $x_{it} = (x_{it}^1, x_{it}^2)$ with $x_{it}^1 = \log(A_j) + u_{it}^1$ and $x_{it}^2 = x_{jt}^1 + \xi_t + u_{it}^2$, then:

$$\log y_{it} = \phi_0 + \phi_a \log A_i + \phi_1 x_{it}^1 + \phi_2 x_{it}^2$$

Claim 2: Higher-order beliefs shocks are equivalent to shocks to equilibrium objects:
Take $x_{it}^1 = \log(A_j) + \varepsilon_{it}^1$ and $x_{it}^h = x_{jt}^{h-1} + u_{it}^h + \xi_t^h,\ \forall h \in \{2, \ldots, H\}$ or $\hat{\mathbb{E}}_{it} \log(y_{jt}) = \log y_{jt}^* + \xi_t$.
$\hookrightarrow$ both lead to $\log(Y_t) = \Phi \bar{a}_t + \psi \bar{\xi}_t$

Rational confusion and irrational sentiments are observationally equivalent

Justification (2): The Townsend (1983) problem

Without a heterogenous prior belief, the recursive solution requires an ever expanding state space to keep track of the infinite regress in forecasting the forecast of other agents (Townsend 1983).

If $y_{i,t} = \phi a_i + \alpha \mathbb{E}_{it}(y_{jt})$, then $y_{i,t} = y^*_{i,t} + \frac{\phi}{1+\alpha} \sum_{k=1}^{\infty} \alpha^k (\xi_t - \mathbb{E}^k_{i,t}[\xi_t])$ with:

$$\mathbb{E}^1_{it}[\xi_t] = \mathbb{E}_{it}[\xi_t],\ \mathbb{E}^2_{it}[\xi_t] = \mathbb{E}_{it}[\mathbb{E}_{jt}[\xi_t]] \text{ and } \mathbb{E}^k_{it}[\xi_t] = \mathbb{E}_{it}\mathbb{E}_{jt}\mathbb{E}^{k-2}_{it}[\xi_t]$$

Huo & Takayama (2023): If the signal follows and ARMA process, then the equilibrium outcome permits a finite ARMA representation, where the coefficients of the lag polynomial are endogenous

$\hookrightarrow$ Fluctuations are sustained by rational confusion which fades away as additional information arrives over time.

Observational equivalence: A heterogenous prior economy can always replicate the equilibrium of a common prior economy and vice-versa, provided fundamental volatility is sufficiently high.

RCE characterization

1. Employment decision given the expected ToT:

$$
\begin{aligned}
V_1(k; z, \xi, K) &= \max_{n} \; V_2(\hat{m}; z, \xi, K) - v(n), \\
\text{s.t.} \quad \hat{m} &= \hat{p}\,\hat{y} + (1-\delta)k, \\
\hat{y} &= z k^{\alpha} n^{1-\alpha}, \\
\hat{p} &= \mathbf{P}(z, \xi, K),
\end{aligned}
$$

2. Capital accumulation:

$$
\begin{aligned}
V_2(m; A, \xi, K) &= \max_{c, k'} \; u(c) + \beta \int V_1(k'; A', \xi', K')\, df(A', \xi' \mid A, \xi), \\
\text{s.t.} \quad c + k' &= m, \\
K' &= \mathbf{G}(A, \xi, K).
\end{aligned}
$$

3. Equilibrium consistency:

$$
\mathbf{P}(z, \xi, K) = \frac{y(z+\xi, z, \xi, K)}{y(z, z, \xi, K)},
$$

$$
\mathbf{G}(A, \xi, K) = \mathbf{g}\Big(\mathbf{y}(A, A, \xi, K) + (1-\delta)K; , A, \xi, K\Big).
$$

State space representation

In log deviations, the equilibrium conditions are (planner's perspective):

$$
\begin{aligned}
\nu n_{it} &= \mathbb{E}_{it}[MRPL_{it}] - \gamma \mathbb{E}_{it}[c_{it}], \\
\gamma\left(\mathbb{E}'_{it}[c_{i,t+1}] - c_{it}\right) &= \left(1 - \beta(1-\delta)\right)\mathbb{E}'_{it}[MRPK_{i,t+1}], \\
p_{it} + y_{it} &= (1-s)c_{it} + s\, I_{it}, \\
y_{it} &= A_t + \alpha k_{it} + (1-\alpha)n_{it}, \\
k_{i,t+1} &= \delta i_{it} + (1-\delta)k_{it}.
\end{aligned}
$$

Where crucially the ToT are given by $p_{it} = Y_t - y_{it}$ and feature in $MRPL = p_{it} + y_{it} - n_{it}$ and $MRPK = p_{it} + y_{it} - k_{it}$

- $\mathcal{F}_{it} = \mathcal{F}_{it_1} = \sigma(z_{it})$ and $\mathcal{F}'_{it} = \mathcal{F}_{it_2} = \sigma(z_{it}, \bar{z}_t, A_t)$. Even though $z_{it} = \bar{z}_t = A_t$, firms are allowed to believe that they may be different from the viewpoint of stage 1.
- Higher-order beliefs come into play only through the expectation of the ToT
- This is equivalent to firms and households expectations of profitability and income

Log-linear solution: a PBE analogy

Reduced form: a dynamic game in which an island's individual best response depends on past outcomes and on expectations of future and current aggregate outcomes.

1. Guess individual policy rules:

$$
\begin{aligned}
n_{it} &= \Lambda_k^n(k_{it} - K_t) + \Lambda_K^n K_t + \Lambda_z^n z_{it} + \Lambda_\xi^n \xi_t \\
c_{it} &= \Gamma_k^c(k_{it} - K_t) + \Gamma_K^c K_t + \Gamma_z^c z_{it} + \Gamma_{\bar{z}}^c \bar{z}_t + \Gamma_a^c A_t + \Gamma_\xi^c \xi_t \\
k_{it+1} &= \Omega_k^k(k_{it} - K_t) + \Omega_K^k K_t + \Omega_z^k z_{it} + \Omega_{\bar{z}}^k \bar{z}_t + \Omega_a^k A_t + \Omega_\xi^k \xi_t
\end{aligned}
$$

2. Solve for the consistency conditions between the individual policy rules and the aggregate outcomes

3. Take $z_{it} = \bar{z}_t = A_t$ and aggregate:

$$
\begin{aligned}
N_t &= \Lambda_K^N K_t + \Lambda_A^N A_t + \Lambda_\xi^N \xi_t \\
C_t &= \Gamma_K^C K_t + \Gamma_A^C A_t + \Gamma_\xi^C \xi_t \\
K_{t+1} &= \Omega_K^K K_t + \Omega_A^K A_t + \Omega_\xi^K \xi_t
\end{aligned}
$$

Solution method: illustration

Take $\alpha = \gamma = 0 \rightarrow$ equilibrium reduces to a static "beauty contest" game with linear BR:

$$n_{it} = \mathbb{E}_{it}[\chi A_t + \omega N_t]$$

Guess $n_{it} = \Lambda_z z_{it} + \lambda_\xi \xi_t$. Therefore $N_t = \Lambda_z \bar{z}_t + \Lambda_\xi \xi_t$ which implies:

$$n_{it} = \mathbb{E}_{it}(\chi A_t + \omega(\Lambda_z \bar{z}_t + \Lambda_\xi \xi_t))$$

Since $\mathbb{E}_{it}(A_t) = z_{it}$ and $\mathbb{E}_{it}(\bar{z}_t) = z_{it} + \xi_t$, then we have:

$$n_{it} = \chi z_{it} + \omega \Lambda_z (z_{it} + \xi_t) + \omega \Lambda_\xi \xi_t$$

Matching coefficients yields

$$\Lambda_z = \frac{\chi}{1-\omega} \quad \text{and} \quad \Lambda_\xi = \frac{\omega}{1-\omega} \Lambda_z$$

Confidence, news and noise

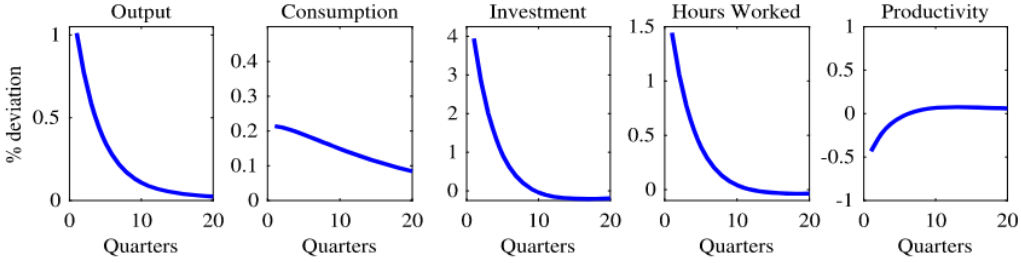


FIGURE 1.—Impulse responses to a positive confidence shock.

A confidence shock causes output, hours, consumption, and investment to move in the same direction. Why?

Confidence, news and noise

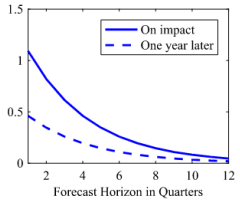


FIGURE 2.—Forecasts of terms of trade, following a confidence shock.

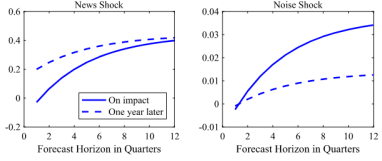


FIGURE 3.—Forecasts of output at different horizons, following a news and a noise shock.

The confidence shock captures expectations of the short term economic outlook:

- Compare with "news shocks":

$$A_t = A_{t-1} + \gamma_{t-1} + \varepsilon_{a,t},\ \gamma_t \sim AR(1)$$
$$\mathcal{F}_t = \sigma(A_t, \gamma_t + \eta_t)$$

- News and noise shocks shift expectations of the medium and long-run prospects

Innovations in ξ_t create broad co-movements of quantity:

$\uparrow \xi_t \Longrightarrow \uparrow$ expected demand $\Longrightarrow \uparrow N_t, K_t \Longrightarrow$ transitory increase to income $\Longrightarrow$ small wealth effect $\Longrightarrow$ transitory $\uparrow Y_t \Longrightarrow C_t, I_t$

Conditionnal moments

CONDITIONAL COMOVEMENTS (6–32 QUARTERS)[a]

	Filtered Data		Our Theory	Alternative Theories			
	(a)	(b)	ξ Shock	I Shock	C Shock	News Shock	E Shock
σ_n/σ_y	1.05	1.29	1.43	1.44	1.44	0.74	0.59
σ_c/σ_y	0.63	0.41	0.25	1.21	1.19	0.35	0.19
σ_i/σ_y	3.35	4.04	3.92	8.93	8.93	5.06	4.26
$\sigma_{y/h}/\sigma_y$	0.45	0.63	0.44	0.51	0.49	0.37	0.43
$\text{corr}(c, y)$	0.86	0.80	0.85	−0.92	−0.94	−0.17	0.65
$\text{corr}(i, y)$	0.94	0.95	0.99	0.98	0.99	0.97	0.99
$\text{corr}(n, y)$	0.91	0.88	0.99	0.98	0.98	0.95	0.99
$\text{corr}(c, n)$	0.80	0.93	0.81	−0.98	−0.99	−0.46	0.51
$\text{corr}(i, n)$	0.86	0.82	0.99	0.99	0.99	0.99	0.99
$\text{corr}(c, i)$	0.73	0.76	0.78	−0.98	−0.98	−0.40	0.55
$\text{corr}(y, y/n)$	0.12	−0.23	−0.96	−0.79	−0.84	0.78	0.97
$\text{corr}(n, y/n)$	−0.31	−0.66	−0.98	−0.91	−0.92	0.56	0.92

[a]Columns (a) and (b) refer to the residuals that obtain, respectively, from the projection of the data on current and past TFP and from the removal of the technology shock identified in the same way as in Galí (1999). All other columns refer to theoretical predictions.

Aggregate demand?

The confidence shock can be interpreted as an Old-Keynesian "aggregate demand shock"
$\hookrightarrow$ does not trigger any inflation-output co-movement.

However: t_1 supply fixed (N_t, K_t), then in t_2 demand adjusts to meet supply through prices
$\hookrightarrow$ not very "Keynesian".

Reverse the timing assumption: consumption and investment fixed in t_1 and employment and output adjust in t_2

$\hookrightarrow$ Same results, because of the short-term nature of the confidence shock, agents don't borrow agains their future income.

Big picture: Information frictions allow to break the NRH without sticky prices. Rational expectations are not unilateraly "classical".

Implementation in a medium-scale DSGE

Bells and whistles: continuum of monopolistically competitive firms on each island, habit formation, investment adjustment costs, various shocks. One version with sticky price and monetary policy, the other without

Estimation: Bayesian ML in the frequency band $\Omega = [6q, 32q]$

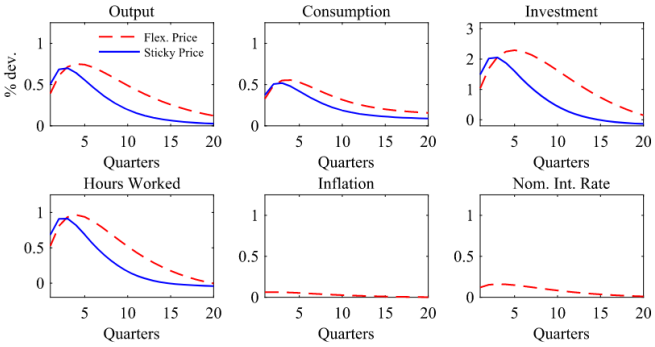


FIGURE 6.—Theoretical IRFs to confidence shock.

Model horse-race

MOMENTS (6-32 QUARTERS)[a]

	Data	FP	SP	SW	Data	FP	SP	SW
		Standard Deviations				*Correlations With Output*		
Y	1.41	1.28	1.36	1.42				
I	5.12	4.46	4.88	4.86	0.94	0.88	0.86	0.74
N	1.56	1.59	1.66	0.97	0.87	0.82	0.83	0.81
C	0.76	0.82	0.91	1.11	0.85	0.78	0.77	0.67
Y/N	0.76	0.91	0.90	0.84	0.07	−0.02	−0.03	0.74
π	0.23	–	0.25	0.34	0.21	–	0.37	0.13
R	0.35	–	0.34	0.35	0.33	–	0.54	0.06
		Correlations With Investment				*Correlations With Hours*		
N	0.82	0.79	0.82	0.67				
C	0.73	0.56	0.47	0.30	0.83	0.65	0.58	0.59
Y/N	0.07	−0.14	−0.21	0.47	−0.43	−0.58	−0.56	0.22
π	0.09	–	0.41	0.18	0.44	–	0.48	0.23
R	0.23	–	0.60	0.23	0.61	–	0.70	0.21

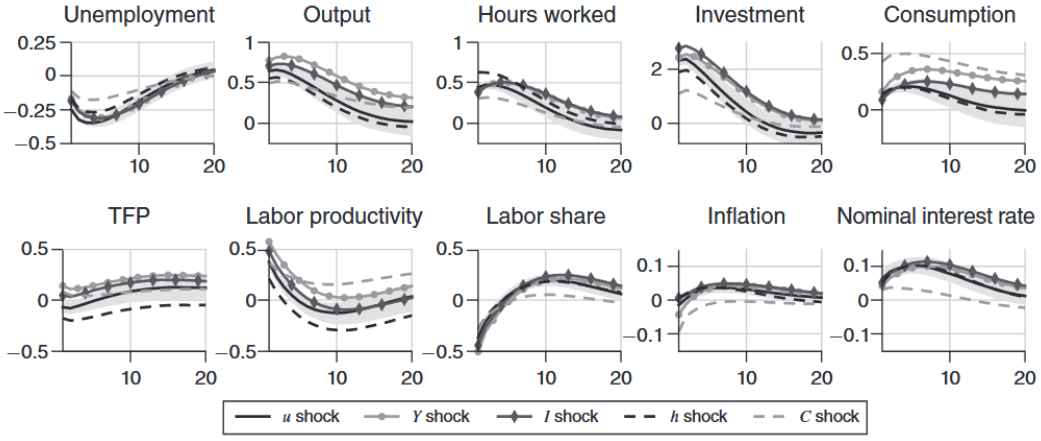
[a]FP and SP stand for our estimated flexible- and sticky-price models, respectively. SW stands for the model in Smets and Wouters (2007).

Model horse-race

POSTERIOR ODDS OF MODEL A VS MODEL B

	Model A	
Model B	Sticky, without	Sticky, with
Flexible prices, without confidence	1.00	1.00
Flexible prices, with confidence	0.36	0.84
Sticky prices, without confidence	–	0.90

Business cycle anatomy



Conclusion

Key point:

- Parsimonious explanation of the business cycle that attributes the deficiency in aggregate demand in part to expectational coordination failure and a lack of confidence.
- Still favored by the data after controlling for a large set of shocks and frictions, including sticky prices and monetary policy.

Going forward:

- Forecasting?
- Co-movements OK, what about variance?
- Robust to the introduction of asset trading?
- Higher-order solution?

La Fin